1BM20 Business Analysis for IT

Assignment C

Requirements Specification

for

Sip&Speak

Version *1.0*

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15/01/2024[1]

[1] *Template based on the book material by Wiegers & Beatty,2013. Permission is granted to use and modify this document.*

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Revision History

Name	Date	Reason For Changes	Version
All team members	05/01/23	First final version	1.0
Charlotte Caris	09/01/23	Revision and comments	1.0
All team members	10/01/23	Resolved comments	1.0
All team members	10/01/23	Second final version	1.0

1. Introduction

Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for the first (1.0) release of Sip&Speak(S&S) application. This document is intended to be used by the developers of the application. Additionally, this document could be used as reference for documentation writers and as a basis for project managers to plan a timeline. All the outlined specifications are dedicated to the release of version 1.0 unless otherwise specified.

Document Conventions

User types and statuses are bolded in tables solely for improving readability. Additionally, assumptions, dependencies, constraints, rules, and requirements are listed in a coded manner to enhance consistency. This code is bolded and constructed with two-to-three letters, followed by a dash and a consecutive number in chronological order. Additionally, the SRS document consistently uses the terms "meet-up" and "S&S moment" interchangeably, both denoting the ultimate in-person meeting. Finally, the reader should pay careful attention to the terminology of user statuses. The use of a continuous or past tense (e.g. connect**ing** or connect**ed**) attribute different enabled functionalities to the user, as described in chapter 2.

Project Scope

The S&S application facilitates language skill exchange among TU/e students by providing the means to connect with one another. Additional details can be found in the Vision and Scope document [1]. This SRS concerns the first release of S&S application and can be seen as a Minimum Viable Product (MVP), which is a basic version of a product with enough features for early users. This first release includes features related to signing up, creating a profile, reviewing other profiles, sending/receiving match requests, starting a connection, finding overlapping time slots, selecting S&S moment location, communicating delays or cancellations with predefined messages, providing feedback to a profile, a messaging tool, removing disruptive users, and generating monthly reports for retailers and TU/e.

Some limitations that are intended to be addressed in future releases are the possibility of editing profiles, supporting other languages than Dutch and English for the application settings, creating a blacklist of for removed users, more structured input fields for the profile details, the possibility to directly search users with a search bar, integration of ice-breaking questions, and synchronizing the S&S moment reservation in the outlook calendar. These features are not required to create a MVP and thus left out of scope. For this release, the scope is limited to a TU/e level.

References

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2. Overall Description

Product Perspective

S&S is an application-based language skill exchange platform. It is an entirely new product because it has a local scope - specifically focused at Eindhoven University of Technology (TU/e) - and offers a unique value proposition that no other competitor has offered yet. S&S is unique in the sense that users are matched based on language proficiency as well as personality. This two-step matching process enables a comfortable setting to practice real-life conversations. This, as well as physically arranged meet-ups at local retailers enable the integration of Dutch and international community. Thus, establishing a greater community feeling.

S&S interacts with several external entities. These are the user classes and four other software systems (Appstore/ Google Play, Retailer's reservation system, LettuceMeet and Amazon Web Services (AWS)). The information flows between S&S and the three other systems are depicted in figure 1, while the user classes are explained in section 2.2.

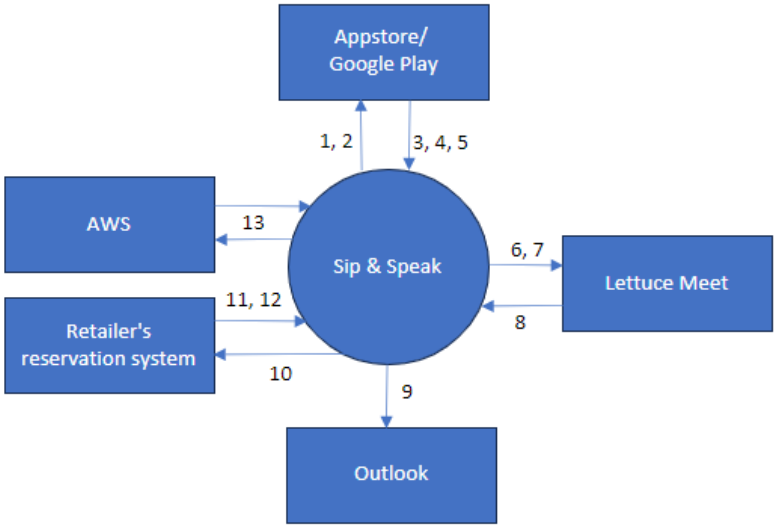


Figure 1: context diagram
(1) App updates: technical updates (i.e. new features) as well as regulation updates (i.e. privacy)
(2) App package: application components required to work on user's devices + app description
(3) Usage and user information report: e.g. number of downloads and aggregated user data
(4) App update approval message
(5) User review and rating of S&S

(6) Meeting-time range (8:00 – 22:00)
(7) E-mail addresses matched users
(8) Filled-in time slots
(9) verification e-mail data
(10) Reservation request
(11) Successful reservation message.
(12) Reservation details: information on the reserved table (e.g. 'table contains an S&S sign)
(13) Cloud data exchange: more information in chapter 2.3

User Classes and Characteristics

According to Wiegers & Beatty (2013), a user class can be identified by the tasks they perform. A distinction should be made between direct users (d) and indirect users (i), where indirect users do not use the system themselves. Nevertheless, they do make use of application's data or services. Therefore, retailers and TU/e are a user class as well because they receive reports. S&S's 'main users' who use S&S as a language exchange application are subcategorized in five types in table 1. This subcategorization depends on different functionalities that users require from S&S. These main users are also the favored users (f), since their satisfaction is most closely aligned with the objectives of S&S. Moreover, disfavored user classes (df) are groups that should not use the application due to safety, legal or security reasons. These are classified as 'disruptive users'.

User class	Description
Registering user (d) (f)	A registering user is new to S&S and has downloaded the application at either the Appstore or Google Play. A registering user signs up and creates a profile.
Matching user (d) (f)	A matching user has successfully created a profile and undergoes the matching process. The **requesting matching user** takes the initiative to review a profile first and send a request to the **receiving matching user**. Users looking for multiple matches become the matching user again.
Connecting user (d) (f)	Matched users that prepare for a physical S&S moment are 'connecting users'. A connecting user fills in a LettuceMeet, picks final time slots and selects a location area. Furthermore, (s)he receives the details about the S&S moment.
Connected user (d) (f)	After meeting up physically, a user becomes a 'connected user'. They can evaluate their experience with the other connected user (the **acquainted connected user**) by leaving a comment or flagging (for major violations).
Repetitive user (d) (f)	A repetitive user is someone who wants to continue the interaction with an acquainted connected user either through meeting up again (through which they become a connected user again) or chatting using the messaging tool.
Disruptive user (df)	A disruptive user should not make use of Sip&Speak due to safety reasons. These users are either socially malicious actors (e.g. people who are disrespectful of other cultures) or intentional liars (e.g. people that do not use the app to learn a new language).
Retailer (i)	Retailers receive usage and user data reports from Sip&Speak such as the number of users. For further details see section 4.3.
TU/e (i)	TU/e receives usage and user data reports from Sip&Speak such as the diversity of its users. For further details see section 4.3.
Admin (d)	The admin is a special user class derived from the project team of Sip&Speak. The admin has the power to review flagged users and decide on their removal.

Table 1: User classes

Registered users have completed the registering process but have not yet started matching: they are idle. The same reasoning holds for 'matched users'.

Operating Environment

OE-1: Hardware platform. S&S's application will be designed as a mobile application that will be exclusively accessible on tablets and smartphones.
OE-2: Operating systems and versions. Since S&S is a mobile application, a Mobile Operating System is needed. Thus, S&S's application will be designed to be compatible with Android versions 7.0 and above, as well as iOS version 10 and above.
OE-3: Geographical location of users. Sip&Speak's application can operate worldwide. However, S&S scheduled meet-ups can only take place in Eindhoven in accordance with the TU/e-scope.
OE-4: Geographical location of servers and databases. S&S's application will make use of the cloud computing services of Amazon Web Services (AWS).
OE-5: Other software components

- LettuceMeet: S&S's application will retrieve the availability list of overlapping time slots form LettuseMeet.
- Outlook: S&Speak's application must be able to send registration verification emails, as well as processing the verification from Outlook.
- Retailer's reservation systems: S&S's application will make an online reservation with the retailer's reservation system for the Sip&Speak-moment.

Design and Implementation Constraints

CO-1: S&S's application must comply with the General Data Protection Regulation [2] in the European Union (e.g. regarding data anonymization and user consent). [Regulatory constraint]
CO-2: S&S's application requires the development team to exclusively use AWS as the primary cloud platform (for servers and databases). [Specific technology and database constraint]
CO-3: S&S's application must be available on both iOS and Android platforms, meaning that developers must maintain separate codebases for iOS and Android. [Development constraint]
CO-4: S&S's application must comply with the regulations of AppStore [3] and Google Play [4]. [Regulatory constraint]
CO-5: S&S's application must adhere to Apple's Human Interface Guidelines [5] for iOS and Google's Material Design [6] for Android. [Development constraint]
CO-6: S&S's application must use LettuceMeet tool for scheduling meet-ups. [Specific tool constraint]
CO-7: S&S's application must support English and Dutch languages. [Design constraint]
CO-8: S&S's application must support a minimum of iOS 10 and Android 7.0 for compatibility with a significant user base. [Development constraint].
CO-9: All textual data used by the application must be stored in the form of XML files. [Specific technology constraint]
CO-10: S&S's application uses only open-source software available under the GNU General Public License [6] for implementing the product. [Specific technology constraint]

Assumptions and Dependencies

AS-1: At least one of the associated retailers has an available spot in the determined time slot.
AS-2: AWS is an affordable cloud service provider.
AS-3: At least 95% of the matched users will fill in the Lettuce Meet within 24 hours of receiving it.
AS-4: 50% of the users will leave feedback on their meet-ups.
AS-5: Associated retailers possess an online reservation system.
AS-6: Online reservation systems can immediately accept or deny a reservation request.

AS-7: The retailer cannot cancel the reservation that is made.

DE-1: The registration of users depends on the effective working of Outlook.
DE-2: The filling in of the LettuceMeet is dependent on accurate input of users.
DE-3: The making of reservations is dependent on the effective functioning of the retailer's online reservation systems.

3. System Features

Use Case List and Diagram

Use case ID-name	*Short description*
UCU-1 **Signing-up**	The **registering user** creates an account in the S&S application by entering the TU/e e-mail. S&S sends a verification e-mail to this address. The registering user must click 'verify' and is automatically directed to creating and confirming a password. Since the e-mail domain name is personalized and owned by TU/e, the TU/e scope is ensured.
UCU-2: **Create the profile**	After signing-up, the **registering user** creates a profile by filling in the information: - Personal details (ch4) - Targeted and offered languages and proficiency levels (ch4) - Biography that better reflects personality aspects (ch4) Then, the registering user saves the profile by clicking on 'Save'. After saving, the profile cannot be edited. As long as the signee does not complete UCU-1 and UCU-2, he is marked an uncompleted user. Otherwise, the signee is marked a 'registered user'.
UCU-4: **Find match**	S&S uses a two-step matching process for matching users. Users find suggestions based on desired language skills and secondly review profiles based on personality matches. The use case is finished by sending a match request.
UCU-5: **Accept/deny request**	**Receiving matching users** can accept or deny a match request. If a request is accepted, two users are matched. Then, a 'match acceptance' notification is sent to both the requesting and receiving matching user. Alternatively, the receiving matching user denies the request. Then, the requesting matching user receives a 'match denial' notification.
UCU-6: **Accept connection**	Two **matched users** can click on 'connect now' in the 'match acceptance' notification or the matchee's profile. After clicking 'connect now', they are **connecting users**.
UCU-7: **Find overlapping time slots**	After clicking 'connect now' by both users, S&S sends a notification both **connecting users** including a LettuceMeet link. Here, they can indicate their availability during the week ahead. Using the filled-in LettuceMeet, S&S can select several overlapping time slots during which both users are available.
UCU-8: **Find S&S moment location**	After S&S has selected several overlapping time slots, S&S creates an overlapping time slot list. Subsequently, S&S sends **connecting users** an availability list containing the overlapping time slot list and associated location *area* preference options (on-campus or off-campus). For each overlapping time slot, connecting users need to confirm their availability and location area preference. Availability is confirmed to ascertain the connecting users are still available. After both users have filled in their preferences, S&S selects the most favorable time and location of the S&S moment (details about 'favorable' can be find in the detailed use case below). Then S&S makes a reservation in the retailer's online system. The use case finishes when S&S sends a notification to both connecting users containing the reservation details.
UCU-9: **Communicate deviations**	If **connecting users** suddenly are running late or are prevented from attending, the hindered user can send a predetermined message (e.g. 'I am five minutes late) and the other user will be notified.

UCU-10: **Place comment**	After the S&S moment has physically taken place, users become **connected**. The **connected user** can leave a comment on the acquainted connected user's profile regarding his or her experience.
UCU-11: **Flag profile**	After the S&S moment, if the connected user experiences a major violation, they can flag the profile of the disruptive acquainted user. The connected user needs to provide an explanation with the reason for flagging.
UCU-12: **Messaging**	After the first S&S moment, **repetitive users** can utilize the messaging tool to chat with acquainted connected users to stay in contact.

Table 2: Use cases for 'user'

An extensive process model can be found in Appendix B detailing the normal use cases for 'user'.

Use case ID-name	**Short description**
UCA-1: **Remove users**	An Admin reviews flagged users based on the flagging explanations. The Admin decides if the situation warrants removal and deletes the profile if necessary.

Table 3: Use cases for 'admin'

Use case ID-name	**Short description**
UCT-1: **Monthly reports**	S&S sends monthly usage and user reports to the TU/e and retailers (ch 4.3).

Table 4: Use cases for 'TU/e' and retailers

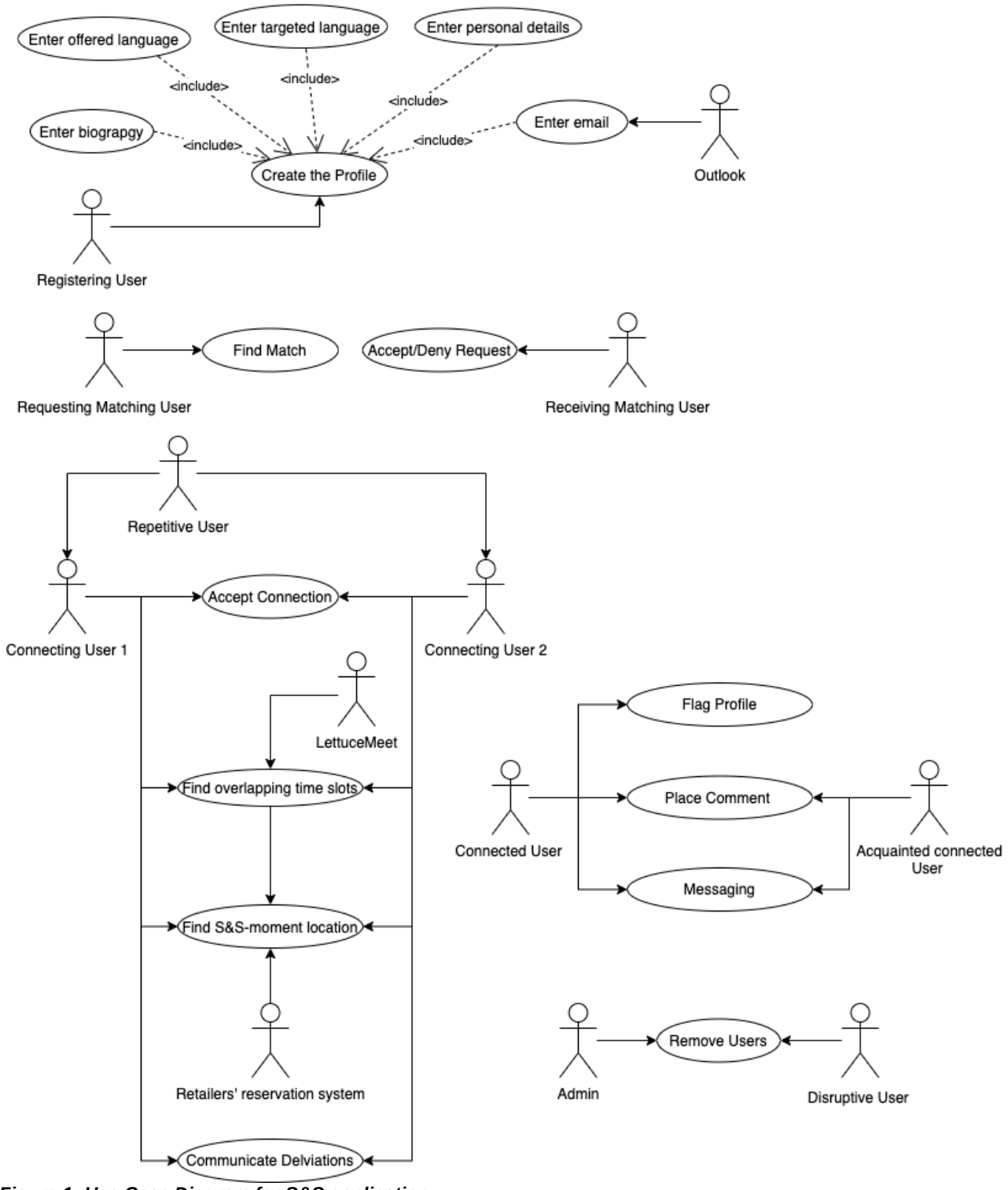


Figure 1: Use Case Diagram for S&S application

Use Cases

UC ID and Name:	**UCU-4 Find match**		
Created By:	X	*Date Created:*	15/12/23
Primary Actor:	Requesting matching user	*Secondary Actors:*	Receiving matching user
Trigger:	The requesting matching user clicks on the 'Find Suggestions' button.		
Description:	Users select "find suggestions" to find language-matched profiles. The user then evaluates the profiles based on the biography on their profile. If a user finds a profile interesting, he or she submits a "send request."		
Precondition	PRE-1. The user has signed up and created a profile (user is marked a registered user).		
Post-conditions:	POST-1. S&S sends the receiving matching users a 'match request' notification.		
Normal Flow:	**UCU-4.0 Sends a request** 1. The requesting matching user views the suggestion list from S&S. 2. The user scrolls through the 'suggestion list'. 3. The user clicks on one of the suggested profiles. 4. The user reviews the profile. 5. The user finds the profile interesting. 6. The user clicks on the "send request" button to send a match request to the receiving matching user.		
Alternative Flows:	**UCU-4.1: Sends multiple requests** "*similar as normal flow but additionally"* 7. The user clicks "back" to be return to 'suggestion list'. 8. The user goes back to '2' of the normal flow.		
Exceptions:	**UCU-4.0.E1 No users to Match** 1. The user receives a 'suggestion failure' notification (S&S cannot find any profiles with the desired language skills). **UCU-4.0.E2 Does not send a request** *"similar as normal flow until"* 5. The user does not find the profile interesting. 6. The clicks "back" to be return to 'suggestion list'. 7. The user clicks "back" to be return to home page.		
Priority:	High		
Frequency of Use:	It is expected that each user will use this use case to search for matches after registering. As receiving requests from other users is also a possibility, it is expected that users will utilize this feature at a very low frequency after their initial use.		
Business Rules:	BR-2 .		
Other Information:	-		
Assumptions:	-		

Table 4: Find match Use Case

UC ID and Name:	**UCU-7 Find overlapping time slots**		
Created By:	X	*Date Created:*	17/12/23
Primary Actor:	Connecting user 1	*Secondary Actors:*	LettuceMeet, Connecting user 2
Trigger:	Both users have clicked "connect now".		
Description:	Connecting users must fill in their available time slots in the generated LettuceMeet meeting for the following week.		
Preconditions:	PRE-1. The receiving matching user has accepted the requests to match.		

Post-conditions:	POST-1. LettuceMeet sends S&S a list with the overlapping time slots.
Normal Flow:	**UCU-7.0 Finding the overlapping time slots** 1. Connecting user 1 and 2 receive the LettuceMeet availability notification. 2. Both users click on the link to the LettuceMeet platform. 3. Both users fill out their available times in the LettuceMeet. 4. Both users confirm their selected choices by saving their choices in the LettuceMeet. 5. LettuceMeet sends a list of the overlapping time slots to S&S.
Alternative Flows:	-
Exceptions:	**UCU-7.0.E1 Not able to enter LettuceMeet** 1. One or both users are not able to enter the platform during the 24 hours. 2. The connection is lost. **UCU-7.0.E2 Did not enter the available time slots** *"similar as normal flow untill"* 3. One or both users did not pick any time slots within 24 hours. 4. The connection is lost. **UCU-7.0.E3 No overlapping time slots** *"similar as normal flow untill"* 5. LettuceMeet sends an empty list to S&S. (There are no overlapping time slots) 6. The connection is lost.
Priority:	High
Frequency of Use:	It is required to perform "find overlapping time slots" once for each pair of users clicking on "connect now." It is expected that an average user is unlikely to schedule many meetings in a single month; hence, this use case will be used at a low frequency.
Business Rules:	BR-3, BR-4, BR-5.
Other Information:	-
Assumptions:	-

Table 5: Find overlapping time slots Use Case

UC ID and Name:	**UCU-8 Find S&S moment location**		
Created By:	X	*Date Created:*	17/11/23
Primary Actor:	Connecting user 1	*Secondary Actors:*	Connecting user 2, Retailer
Trigger:	Connecting user 1 and 2 receive a 'successful availability match' notification.		
Description:	The 'successful availability match' notification includes the overlapping time slot list retrieved from their filled-in Lettuce Meet calendar. They both need to confirm their availability for the overlapping time slots and need to select the preferred option between "on-campus" and "off-campus". Then S&S schedules the S&S moment and sends the users a 'successful meet-up' notification containing the details of S&S moment.		
Preconditions:	PRE-1. There is at least one overlapping timeslot retrieved from the filled-in Lettuce Meet calendar of both connecting users.		
Post-conditions:	POST-1. S&S schedules an S&S moment aligned with both users' preference. POST-2. S&S schedules an S&S moment aligned with one users' location preference.		
Normal Flow:	**UCU-8.0 Selected S&S moment aligned with user preferences** 1. Connecting user 1 and 2 open the 'successful availability match' notification 2. Both users confirm the time slots they are available. 3. For each selected time, both users select either "on-campus" or "off-campus" or both choices, as their location area preference. 4. Both users confirm their choices by clicking on "save". 5. A common time & location preference exists. S&S picks a retailer based on a common location area preference for an overlapping time slot. S&S makes a reservation for the		

	earliest available time for that common location area preference in the retailer's online reservation system. 6. Both connecting users receive a 'successful meet-up' notification.
Alternative Flows:	**UCU-8.1 Selected S&S moment not aligned with user preferences** "*similar as normal flow untill"* 5. A common time exists but no common location preference exists. S&S reserves a retailer location on campus for the earliest common time slot. 6. A 'successful meet-up' notification is sent to both users containing the details of S&S moment.
Exceptions:	**UCU-8.0.E1 Did not enter the time and location** *"similar as normal flow"* 2. One or both did not confirm time slots nor picked a location area within 24 hours. 3.The connection is lost. 4. Both users are sent an 'unsuccessful meet-up' notification. **UCU-8.0.E2 No overlapping timeslot exits** *"similar as normal flow"* 4. No common time exists between users choices. 5.The connection is lost. 6. Both users are sent an 'unsuccessful meet-up' notification.
Priority:	High
Frequency of Use:	Successful "find overlapping time" leads to this use case. Hence, for every successful "find overlapping time slots," "find S&S moment location" needs to be performed, which results in a low frequency of use for each user.
Business Rules:	BR-6, BR-7, BR-8.
Other Information:	-
Assumptions:	AS-1: At least 95% of the matched users will fill in the Lettuce Meet within 24 hours of receiving it. AS-2: S&S's application can make an online reservation with the retailer's online reservation system for the S&S moment, and the system will immediately accept or deny a reservation request. AS-3: At least one of the associated retailers has an available spot in the determined time slot, and the retailer cannot cancel the reservation that is made.

Table 6: Find S&S moment location Use Case

UC ID and Name:	**UCA-1 Remove users**		
Created By:	X	*Date Created:*	15/12/23
Primary Actor:	Admin	*Secondary Actors:*	Disruptive User
Trigger:	A connected user flags a disruptive acquainted connected user.		
Description:	Admin reviews the reason for flagging, potential previous explanations for flagging and comments on the profile. Based on this, the admin decides on removing the disruptive user.		
Preconditions:	PRE-1. S&S moment (physical meet-up) took place.		
Post-conditions:	POST-1. The disruptive user cannot use S&S anymore.		
Normal Flow:	**UCA-1.0 Removing a user** 1. Admin receives a flagging alert. 2. Admin reviews the flagged user's feedback (flagging explanations and profile comments) 3. Admin decides to remove the user.		

	4. Admin deletes the user's profile.
Alternative Flows:	**-**
Exceptions:	**UCA-1.0.E1 Not removing a user** *"similar as normal flow until"* 3. Admin decides not to remove the user.
Priority:	Medium
Frequency of Use:	This use case is not expected to be performed often, and admin is not expected to handle many flagging issues.
Business Rules:	BR-11.
Other Information:	-
Assumptions:	-

Table 7: Remove user Use Case

Business Rules

ID	Rule Definition	Type of Rule	Static or Dynamic	Source
BR-1	Only TU/e students can register in S&S.	Constraint	Static	User
BR-2	Users need to have completed their registration before undergoing the matching process.	Constraint	Static	User
BR-3	The selected available time slots in the LettuceMeet should at least be 30 minutes to ensure the retailers' profit.	Constraint	Dynamic	User
BR-4	Within 24 hours of clicking 'connect now', users need to enter their availability.	Constraint	Dynamic	User
BR-5	If "pick timeslot" fails, the connection is 'lost' which means that both users have to click 'connect now' again if they want to schedule a meeting.	Constraint	Static	User
BR-6	If at the stage of picking the location, no match was found between options on the ground of location, the earliest selected timeslot is selected and the location is set to "on-campus".	Constraint	Static	User
BR-7	Within one day of sending the time and location area list, users need to enter their preference.	Constraint	Dynamic	User
BR-8	If "pick location area", the connection is 'lost' which means that both users have to click 'connect now' again if they want to schedule a meeting.	Constraint	Static	User
BR-9	A user can only place comments on profiles or flag profiles of acquainted connected users.	Constraint	Static	Admin
BR-10	Messaging feature will only be available to users after their first physical S&S moment.	Constraint	Static	User
BR-11	Only a verified individual from the S&S team can be the Admin.	Fact	Static	Admin

Table 8: Business rules

4. Data Requirements

Logical Data Model

The entity-relationship diagram is shown in figure 2. To decrease cluttering of the model, the associated attributes are listed as data elements in chapter 4.2. The different user classes are

marked yellow for clarity. The E-R model follows the enhanced conventions of the ER model. The following links can be used by the reader if (s)he is unfamiliar with these conventions: 1, 2 , 3.

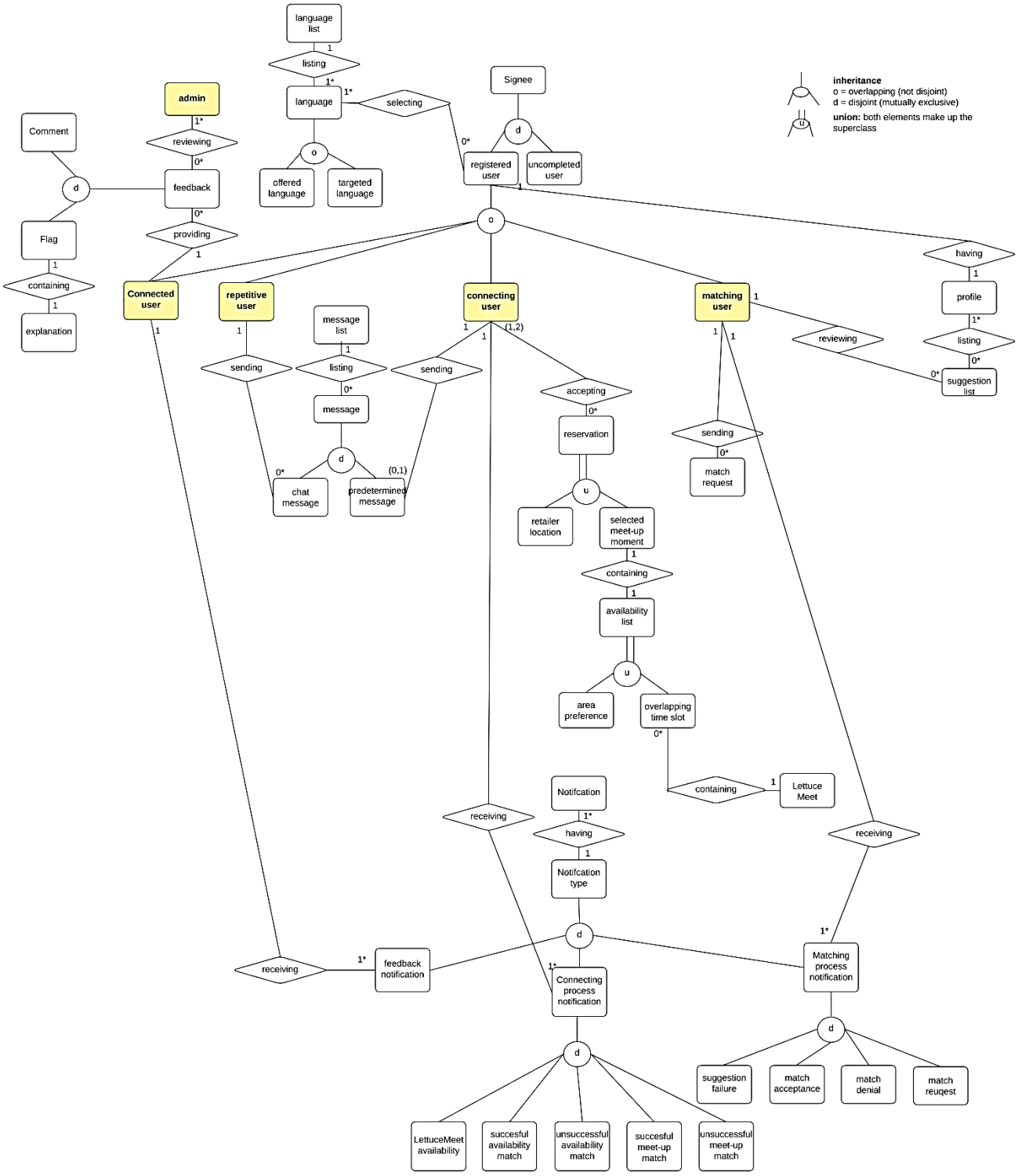


Figure 2: E-R model

Data Dictionary

Although it is recommended to alphabetically organize data elements in a data dictionary, an exception has been made to enhance readability of the data entities depicted in the ER data model of section 4.1. The following data dictionary is then expressed in terms of attributes of the respective data entities.

Data entity: Signee				
Data Element	**Description**	**Composition or Data Type**	**Length**	**Values**
Signee ID	Unique identifier for a signee	Alphanumeric	8	System-generated sequential code. The first ID is S0000001.
TU/e email	Email with which the signee signs in. Exclusive of domain "@student.tue.nl"	Alphanumeric	60	Periods permitted. The TU/e email address must have the domain: @student.tue.nl
Password	The password with which the signee creates an account.	Alphanumeric	20	Symbols permitted.
Registration date	The date on which the signee registered	MM/DD/YYYY	10	
Data entity: Registered user				
Registered user ID	Unique identifier for a registered user	Alphanumeric	8	System-generated sequential code. First ID is RU000001.
Profile	The profile details of the registered user	Registered user ID + Profile ID + Profile picture + Username + Offered Languages + Targeted Languages + Biography + (Comment)		
Registered user	A signee that has a complete profile	Profile + Profile completion date		
Data entity: Profile				
Profile ID	Unique identifier for the profile of a registered user	Alphanumeric	8	System-generated sequential code. First ID is P0000001.
Profile picture	A picture that the user uploads	JPG		
Profile completion date	The date in which the registered user completed his/her profile	MM/DD/YYYY	10	
Username	Username built using the name and last name	First name + Last name		
First name	First name of the registered user	Alphabetic	30	
Last name	Last name of the registered user	Alphabetic	30	
Age	The age of the user	Integer	2	
Study	The studies that the user follows	Alphabetic	80	

Introduction	Brief text that the user provides for introducing his/her-self	Alphanumeric	250	Symbols permitted.
Hobby	A hobby that the user has	Alphanumeric	80	Symbols permitted.
Conversation topics	Three conversation topics	Alphanumeric	250	Symbols permitted.
Biography	Complements the profile details of the user	Age + Study + Introduction + Hobby + Conversation topics		
Data entity: Languages				
Language	Official name of a language	Alphabetic	30	All languages listed in ISO 639-2
Languages list	All languages listed in ISO 639-2	1:m{Language}		
Proficiency	Level of proficiency of a language	Alphanumeric	2	A1, A2, B1, B2, C1, C2
Offered language details	The language that a user speaks fluently	1:1{Languages list} + Proficiency		
Targeted language details	The language that a user wants to improve	1:1{Languages list} + Proficiency		
Offered Languages	The languages that a user speaks fluently	1:n{Offered language details}		
Targeted Languages	The languages that a user wants to improve	1:n{Targeted language details}		
Data entity: Suggestion list				
Suggestion list ID	Unique identifier for a signee.	Alphanumeric	8	System-generated sequential code. The first ID is SL000001.
Suggestion list	A list of alike profiles.	0:10{Alike profile}		
Alike profile	True if a profile has at least one offered language as the target language of the user and vice-versa.	Boolean	450	True or false for targeted language matching offered language and vice-versa.
Data entity: Match request				
Match request ID	Unique identifier for a match request.	Alphanumeric	8	System-generated sequential code. The first ID is MR000001.
Requesting matching user	The registered user ID that sends a match request.	Registered user ID		
Receiving matching user	The registered user ID that receives a match request.	Registered user ID		
Match request date	The date on which the match request was sent.	MM/DD/YYYY	10	
Data entity: Notification				
Notification type	The type of notification a registered user can receive.	Alphabetic	31	Matching process notification type, Connecting process notification type, Feedback notification type.
Type of matching notification	Type of notification a matching user can receive.	Alphabetic	18	Suggestion failure, match acceptance,

				match denial, match request.
Type of connecting notification	Type of notification a connecting user can receive.	Alphabetic	29	LettuceMeet availability, successful availability match, unsuccessful availability match, successful meet-up match, unsuccessful meet-up match.
Feedback notification	Type of notification a repetitive user can receive	Comment + date of comment		
Data entity: Message				
Message ID	Unique identifier for the first Message	Alphanumeric	8	System-generated sequential code. The first ID is ME000001.
Message	Two types of messages.	Alphabetic	21	Chat message, Predetermined message
Chat message	A text box where a message can be written to chat with a user with whom a connected ID has been generated.	Alphanumeric	450	
Predetermined message	A predetermined message that the connecting user can send to his connecting partner.	Alphanumeric	39	'I am running late 5 min', 'I am running late 10 min', 'I am running late 15 min', 'I need to cancel due to transport issues', 'I got sick sorry', 'I won't make it I am very sorry'.
List of messages	A list with the messages received per user	Registered user ID + Message ID		
Data entity: Feedback				
Feedback ID	Unique identifier for feedback.	Alphanumeric	*8*	System-generated sequential code. The first ID is F0000001.
Comment	A text box where a comment with the experience of the meet-up can be written.	Alphanumeric	250	Symbols permitted.
Date of comment	The date on which the comment was given.	MM/DD/YYYY	10	
Flag explanation	A text box where an explanation for flagging must be written.	Alphanumeric	450	Symbols permitted.
Date of flag explanation	The date on which the flag explanation was given.	MM/DD/YYYY	10	
Data entity: Retailer location				
Off-campus retailer name	The name of the associated retailer for the meet-up	Alphanumeric	80	Hyphens and commas are permitted.
Off-campus retailer location	Street and house number of the retailer.	Alphanumeric	80	Hyphens and commas are permitted.
Off-campus retailer	The off-campus retailer name and the off-campus retailer location	Off-campus retailer name + Off-campus retailer location		Hyphens and commas are permitted.

On-campus retailer name	The name of the associated retailer for the meet-up	Alphanumeric	80	Hyphens and commas are permitted.
On-campus retailer location	Building name, floor, and number	Alphanumeric	80	Hyphens and commas are permitted.
On-campus retailer	The on-campus retailer name and the on-campus retailer location	On-campus retailer name + On-campus retailer location		Hyphens and commas are permitted.
Data entity: Selected meet-up moment				
Overlapping time slot	Overlapped time slot retrieved from LettuceMeet.	Date, DD/MM/YYYY, Time, HH:MM	25	
Availability list	List of overlapped time slots.	0:m{Overlapping time slot} + Area preference		
Selected time slot	The time slot for the meet-up.	Date, DD/MM/YYYY, Time, HH:MM + Area preference	35	
Area preference	The location preference for each time slot in the availability list.	Alphabetic	10	Off-campus, On-campus
Data entity: Reservation				
Reservation ID	Unique identifier for every reservation case	Alphanumeric	8	System-generated sequential code, beginning with 1. The first ID is R0000001.
Reservation details	The data and time of the selected time slot plus the name and location of the retailer	Selected time slot + off or on-campus retailer		
Case identifiers: To keep track of usage for reports (see 4.3)				
Connected ID	Unique identifier for every connected case	Alphanumeric	8	System-generated sequential code, beginning with 1. The first ID is C0000001.
Matched ID	Unique identifier for every matched case	Alphanumeric	8	System-generated sequential code, beginning with 1. The first ID is M0000001.
Deletion ID	Unique identifier for every deleted account case	Alphanumeric	8	System-generated sequential code, beginning with 1. The first ID is D0000001.

Reports

S&S generates a report with usage and user data for retailers as well as TU/e. Both reports will contain a 'user engagement' section. Nevertheless, the contents of this section differ between retailers and TU/e because they derive different value from associating with S&S. For retailers this is an increased visibility and sales resulting from actual meet-ups, while for TU/e the main value is the increased university atmosphere and image resulting from the integration of the Dutch and international community. Thus, for both data regarding the application's reach is relevant. This regards the number of users (created profiles), number of matched users, frequency of meet-ups assigned at which location and number of cancelled meet-ups. Additionally, for TU/e information

regarding the number of repetitive users that use the messaging tool to follow up on their meeting is relevant. This is measured through: the number of used messaging channels and the average number of messages send by repetitive users.

The second section is about 'user profile', displayed anonymously (cannot be traced back to a specific person). This regards the statistical display of the associated languages, ages, studies, nationalities, and genders. This information is provided to both for insights in the associated users.

Additionally, TU/e will receive a report section concerning 'feedback'. In this way the university is enabled to reflect on the prospected positive effects on the community feeling and notified about potential unsafe engagements. The latter would impact S&S's reputation but is required to be shared with TU/e to secure their funding. The 'feedback' section entails: number of flagged users, reasons for flagging statistically organized, number of deleted accounts, number of comments and the nature of comments statistically organized. 'Statistically organized' means that all comments and reasons for flagging will be categorized and presented proportionally. The first categories are predefined and will be adjusted later in the process depending on the feedback provided. The categories for flagging are: disrespectful against different cultures, sexist behavior, inappropriate intentions of application usage (e.g. usage for dating purposes), refusal to practice language skills, refusal to pay for consumptions and other inappropriate behavior.

Naturally, far more user and usage data can be obtained. It is determined that in the initial release scope the provision of this data should satisfy both retailers and TU/e.

Data Acquisition, Integrity, Retention, and Disposal

DI-1: S&S shall remove user profiles that have been inactive for 1 year.
DI-2: S&S shall retain user profile data, including matched connections and messaging history, for 2 years after the last interaction day.
DI-3: In line with the AVG law regulations, Sip&Speak shall ensure the integrity of user-generated content such as profile information all the time.
DI-4: Upon user request or account deletion, Sip&Speak shall dispose of all associated data, including user profiles and messages within 60 days.

5. External Interface Requirements

User Interfaces

UI-1: S&S shall provide the sign-up page as the first interface including the elements defined in Appendix C, UI-1.
UI-2: S&S shall have a home page interface including the elements as defined in Appendix C, UI-2.
UI-3: S&S shall show the action-buttons associated with the match-status on the user's profile as defined in Appendix C, UI-3.
UI-4: When the user clicks 'find suggestions', S&S shall display a list of suggestions.
UI-5: S&S shall provide a list of new and previous notifications to users in the notification interface.
UI-6: S&S shall provide the generated link to LettuceMeet in user profile and related notification message.
UI-7: S&S shall provide a list of overlapping time slots to users to pick their location area preference.
UI-8.1: S&S shall provide a predetermined messaging interface for connecting users.

UI-8.2: S&S shall provide an open-chat messaging interface for connected users.
UI-9: After two users are connected, S&S shall provide the commenting and flagging interface.

Details of each user interface can be consulted in Appendix C.

The prototype is available at this link: https://youtu.be/EdSvwMKFsZU

Software Interfaces

SI-1: S&S application shall be uploaded in Playstore/ Appstore as well as corresponding updates.
SI-2: LettuceMeet shall send the two connecting users a calendar link spanning 7 days (starting the day after the latest "connect now") from 8 AM to 9 PM.
SI-3: The connecting users shall be able to input their availability in the LettuceMeet calendar.
SI-4: LettuceMeet shall return a list of overlapping time slots to S&S.
SI-5: S&S application shall make a reservation in the retailer's online reservation system.
SI-6: S&S shall use the cloud computing services of Amazon Web Services for business applications as defined by AWS.
SI-7: S&S application shall integrate a native image picker library to enable users to upload their profile picture.

Hardware Interfaces

No hardware interfaces have been identified that interface with the S&S application.

Communications Interfaces

CI-1: S&S application shall communicate with Outlook via a unique verification link sent to the user's Outlook email.

CI-2.1: S&S shall send a usage and user data report to TU/e every month in XLS format according to chapter 4.3.
CI-2.2: S&S shall send a usage and user data report to associate retailers every month in XLS format according to chapter 4.3.
CI-3: S&S shall provide real-time notifications to users according to the description in Appendix C.

6. Quality Attributes

Usability

USE-1: 95% of the signees shall be able to complete their profiles within 20 minutes.
USE-2: The user shall be able to switch between profiles in the suggestion list with an error rate of 1 mistake every 3 profiles (e.g. accidentally going to homepage instead of to suggestion list).
USE-3: A person who has never used S&S before shall be able to pick a time slot* with no more than 5 minutes of orientation.

* Same for: 'pick location area', 'communicate delays', 'comment on profile', 'flag profile', 'message'.

USE-4: After the first physical meet-up (S&S moment), the user shall understand which functionalities his/her status with another user (matching, connecting, connected or repetitive) allows him/her to perform without needing help.
USE-5: The admin shall be able to remove a flagged user within 5 minutes.
USE-6: S&S interfaces shall follow the 10 design principles of usability from Jakob Nielsen [7].

Performance

PER-1: S&S application shall send the verification email within a maximum of 1 minute after the user has signed in.
PER-2: S&S application shall display notifications to users within a maximum of 3 seconds after the user provides information to S&S.
PER-3: S&S application shall orchestrate a physical meet-up moment (S&S- moment) for 90% of the connecting users within three weeks of completing their registration.
PER-4: S&S application shall enable sending the chat message within a maximum of 3 seconds after the user clicks send.
PER-5: Removal of a disruptive user's account shall take no more than 5 seconds after approval of the admin.
PER-6: S&S application shall handle up to 100 concurrent users without significant performance degradation.

Security

SEC-1: All messages* shall be encrypted with 256-bit encryption.

* Passwords, TU/e-emails, suggestion lists, notifications, availability lists, arranged locations and times for meet-ups, associated retailer lists

SEC-2: Only admins shall be permitted to access information regarding flagged users, per BR-15.
SEC-3: S&S application shall lock a user's account after three unsuccessful successive attempts within a period of 10 minutes.

SEC-4: S&S application shall conform to the ISO/ IEC 27001 Information Security Management System Security [8] standard.

Safety

SAF-1: S&S application shall permit the messaging tool only to connected users.
SAF-2: S&S application shall verify each registered e-mail address to belong to the TU/e student that initiated the registration process.
SAF-3: The admin shall remove any user who has been flagged three times by three different users.

Other

INS-1: Installability. Users shall be able to download S&S application within maximum 2 minutes.
INT-1: Interoperability. The system shall be able to export XLS files with the required information specified in section 4.3 (reports).
AVA-1: Availability. S&S application shall be available at least 99% of the time between 6:00 A.M. and 1:00 A.M. according to the Dutch Time Zone.
AVA -2: Availability. S&S application shall be available at least 90% of the time between 1:00 A.M. and 6:00 A.M. according to the Dutch Time Zone.

7. Internationalization and Localization Requirements

IL-1: The S&S application content is available in both American English and Dutch.
IL-2: All text in user interfaces, excluding profile comments and other input texts from the users, shall be translated into the selected language.
IL-3: S&S shall adjust messages' timestamps to the users' mobile time zone.
IL-4: S&S moment scheduling activities shall adhere to the central European time zone.
IL-5: Standard dates within the app shall be presented in DD/MM/YYYY format following the Gregorian calendar.
IL-6: The time format utilized in the S&S application shall be HH:MM.
IL-7: The Western order for first names and last names shall be used in usernames and profiles.

8. Other Requirements

All relevant requirements are defined in previous sections.

Appendix A: Glossary

LettuceMeet: a scheduling tool where users can fill in their availability in a predetermined time range. Through the following link the tool can be accessed to enter one's availability for a week: https://lettucemeet.com/l/qgpE3

User Status/ Relationship Stages:

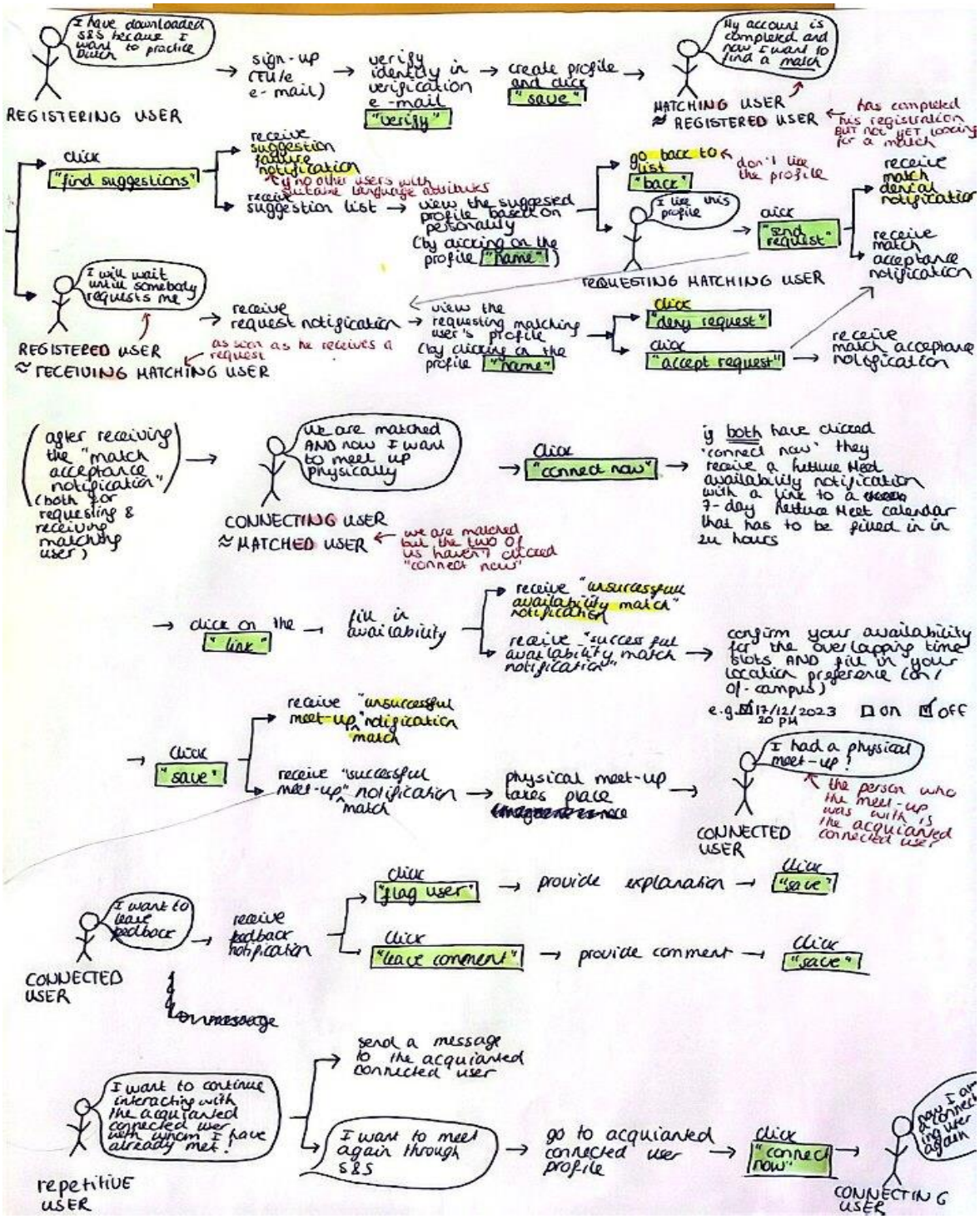
- **Matching:** The state where two profiles match based on the language selection. The matching user sends a match request to the receiving matching user, but he has not responded yet. In this state, these two users are matching.
- **Requesting matching user:** The user who eventually sends a match request to a user whose language selection matches. This user initiates the matching process by clicking 'find suggestions' and reviewing the profiles.
- **Receiving matching user:** The user who receives a match request and can accept or deny the request. The user has not accepted or denied the request yet.
- **Matched:** After the match request has been accepted by the receiving matching request, two users are matched.
- **Matched users:** Two users who have sent and accepted the match request.
- **Connecting:** After two users get matched, they click on the "connect now" button to schedule a S&S moment, and then they are connecting.
- **Connecting users:** Two users who are in the process of scheduling a S&S moment, which can be clicked on "connect now," received the LettuceMeet link, filled in LettuceMeet, etc. Note that a connected user again become a connecting user if they enter the process of scheduling a new S&S moment.
- **Connected:** After the first S&S moment, two connecting users will get connected.
- **Connected users:** After the first S&S moment, two connecting users will become connected users.

S&S: Shorter name for Sip&Speak.

S&S-moment: the physical meet-up moment for which users agree on day and time for practicing a language.

Appendix B: Analysis Models

The process model presented here follows no standard modelling conventions and is simply drawn to increase understanding of the reader as to which actions a user with a certain status (e.g. matching) can perform. Buttons are marked green and notifications yellow. A user state is marked with a person-icon. Lastly, only the 'happy-path' is drawn fully.



Appendix C: User interface

ID	Name	Elements of each UI	Functionality	Link to other UI
UI-1	Sign-up page 1	Enter email	Text box for entering the email address.	-
		Verify	Button that upon pressing, after filling in the email, the app will check your email.	UI-1.1
UI-1.1	Waiting for verification	-	This message is shown till email is verified in the verification e-mail sent by S&S.	UI-1.2
UI-1.2	Sign-up page 2	Password	Text box for entering the password.	-
		Re-Password	Text box for re-entering password to confirm.	-
		Continue	Button that upon pressing, after filling the boxes, you are directed to the profile creation	UI-1.3
UI-1.3	Creating profile	Profile picture	Button to upload user's picture (more in chapter 5.2 SI-6).	Native image selection popover
		First name	Text box for entering the personal details.	-
		Last name		
		Offered language	Drop down list containing languages. After selecting a language, proficiency (e.g. A2) must be selected from the list as well.	-
		Targeted language		-
		Age	Text box for entering the other asked details.	-
		Study		-
		Introduction		-
		Hobby		-
		Conversation topic		-
		Save	Button to save the created profile.	UI-2
UI-2	Home page	Matched users	A list of users that are being differentiated between their type (matched, connecting, connected).	UI-3.4
		Connecting users		To the profile
		Connected users		UI-2.4
		Settings	Button to adjust the language setting of the application.	UI-2.1
		Find suggestions	Button that directs to the suggestion list.	UI-2.2
		Notifications	Button that directs to the notification list.	UI-2.3
		My profile	Button that directs to one's own user profile.	UI-3.1
UI-2.1	Language settings	English	A list where users can indicate the language setting of the app.	-
		Dutch		-
		Save	Button to save the selected language and then the user is directed back to the home page.	UI-2
		Back	Button that directs back to the home page.	UI-2

UI-2.2	Suggestion list	Profiles	A list of suggested profiles for the user. Upon clicking on one profile, user is directed to the complete profile.	UI-3.2
		Back	This button takes the user back to the home page.	UI-2
UI-2.3	Notification list	Suggestion failure	Description: 'No profiles found, try again later.' Button: <Back> : (UI-2.3)	-
		Match request	Description: 'You have a request from <username>.' Button: <go to profile>: (UI-3) <Back> : (UI-2.3)	-
		Match denial	Description: 'Your request was denied.' Button: <Back> : (UI-2.3)	-
		Match acceptance	Description: 'You are now matched with <username>, let's start connecting.' Button: <connect now>: (UI-3) <Back> : (UI-2.3)	-
		LettuceMeet availability	Description: 'Fill in your availability.' Button: <LettuceMeet link>: (Directed to LettuceMeet) <Back> : (UI-2.3)	-
		Successful availability match	Description: 'Confirm your availability in the overlapping time slots and select your location area preference.' Button: <Meet-up preference>: (UI-4) <Back> : (UI-2.3)	-
		Unsuccessful availability match	Description: 'No overlapping time slots found, try again later' Button: <Back> : (UI-2.3)	-
		Successful meet-up match	Description: 'Your S&S moment is scheduled for <date & time> at <retailer location>.' Button: <Back> : (UI-2.3)	-
		Unsuccessful meet-up match	Description: 'After confirmation, no suitable moment found' Button: <Back> : (UI-2.3)	-
		Feedback	Description: 'Leave feedback for <username>' Button: <Feedback>: (UI-3.2) <Back> : (UI-2.3)	-
		Back	This button takes the user back to the home page.	UI-2
UI-3	Profile	Picture	Display of the personal details of the user.	-
		Username		-
		First name		-
		Last name		-
		Offered language		-
		Targeted language		-
		Age		-
		Study		-

		Biography		-
		Comments		-
		Send request	[Enabled when matching ('find suggestions' trigger button) and disabled after matched] Button which the user can use to send a match request.	-
		Accept match	[Enabled after a match request is sent and disabled after matched] Button which the user can accept the match request.	-
		Deny match	[Enabled after a match request is sent and disabled after matched] Button which the user can deny the match request.	-
		Connect now	[Enabled after matched and disabled while connecting] Button to send a new connection request.	-
		LettuceMeet	[Enabled after both users clicked on "connect now"] Button to direct the user to LettuceMeet.	To Lettuce-Meet
		Meet-up preference	[Enabled after both users filled in the LettuceMeet] Button to direct the user to overlapping time list.	UI-4
		Message	[Enabled after filling in the location preference] Button that upon clicking, user will be directed to a chat environment.	UI-3.1
		Feedback	[Enabled after connected] Button to be directed to the "feedback" interface.	UI-3.2
		Back	This button takes the user back to the home page.	UI-2
UI-3.1	Chat	Reading messages	A page to read the previously sent and received messages.	-
		Writing a message	[Enables after connected] A text box to write a message.	-
		Predetermined message	[Enables after filling in the location preference and before connected] A predetermined message to send to connecting user.	-
		Send	Button to send the previously written/ chose message.	-
		Back	Button to take the user back to the home page.	UI-2
UI-3.2	Feedback	Write a comment	Text box to write a comment about the user.	-
		Flag the user	A text box for writing the reason for the flagging.	-
		Save	Button to save the comment or flag and take the user back to the home page.	UI-2
		Back	Button to take the user back to the home page.	UI-2
UI-4	Meet-up location preference	Time slots	A list of the overlapping time slots that upon selecting is confirmed.	-
		Choose area preferences	For each confirmation of the time slots, the user need to pick between the off-campus or on-campus preference. The user also can choose both options for each timeslot by choosing off-campus and on-campus.	-
		Save	Button to save the choices and take the user back to the home page.	UI-2
		Back	Button to take the user back to the home page.	UI-2

Table 9: User interfaces details